Roll No:													

BTECH
(SEM. VI) THEORY EXAMINATION 2025-26
DATA ANALYTICS

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Qno.	Question	CO	Level
a.	Explain structured vs semi-structured data.	1	K2
b.	List the phases of the Data Analytics Lifecycle.	1	K1
c.	Define multivariate analysis.	2	K1
d.	Define Bayesian inference.	2	K1
e.	What is stream data? Give one example.	3	K1
f.	What is the Apriori algorithm?	4	K2
g.	What are NoSQL databases?	5	K2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

Qno.	Question	CO	Level
a.	Explain the Data Analytics Lifecycle in detail with a neat diagram.	1	K2
b.	Explain the linear regression model along with its assumptions. Using the least squares method, fit a regression line for the following data and obtain the equation of the line ($y = mx + c$). $Data$: (1,2), (2,4), (3,5), (4,4), (5,5)	2	K3
c.	Discuss the concept of filtering data stream processing. Explain Bloom filtering in detail.	3	K2
d.	Provide an in-depth comparison between the CLIQUE and ProCLUS clustering algorithms. How do these methods handle challenges such as noise, outliers, and varying cluster shapes?	4	K3
e.	Discuss the components of Hadoop ecosystem: Apache Hive, Apache Pig, Apache HBase.	5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

Qno.	Question	CO	Level
a.	What happens in the 'model planning' and 'model building' stages? How are these two different?	1	K2
b.	Discuss the evolution of Big Data and need for data analytics in modern organizations.	1	K2

4. Attempt any *one* part of the following: **07 x 1 = 07**

Qno.	Question	CO	Level
a.	Explain neural networks including learning, generalization, and competitive learning.	2	K2
b.	Discuss Fuzzy membership function. Apply Union, Intersection, Complement, Bold Union and Bold Intersection operations on fuzzy sets A and B.	2	K3

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	Given: A = {(10,0.2), (20,0.4), (25,0.7), (30,0.9), (40,1.0), (50,0.4)} B = {(10,0.4), (20,0.1), (25,0.9), (30,0.2), (40,0.6), (50,0.6)}		

5. Attempt any *one* part of the following: **07 x 1 = 07**

Qno.	Question	CO	Level
a.	Discuss Real-Time Analytics Platforms (RTAP) and explain applications like sentiment analysis and stock prediction.	3	K2
b.	Apply the Flajolet–Martin algorithm on the following data stream to estimate the number of unique elements: $S = \{1,3,2,1,2,3,4,3,1,2,3,1\}$ Given the hash function: $h(x) = (3x + 1)\ mod\ 8$ Compute the hash values for each element i. Represent the hash values in 3-bit binary form ii. Determine the maximum number of trailing zeros (R) iii. Estimate the number of distinct elements using 2^R	3	K3

6. Attempt any *one* part of the following: **07 x 1 = 07**

Qno.	Question	CO	Level
a.	Discuss market basket analysis and association rule mining.	4	K2
b.	Cluster the following eight points (with (x, y) representing locations) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9) • Initial cluster centers are: A1(2, 10), A4(5,8) and A7(1, 2). • The distance function between two points a = (x1, y1) and b = (x2, y2) is defined as $d(a,b) = \|x2 - x1\| + \|y2 - y1\|$ Use K-Means Algorithm to find the three cluster centers after implanting all eight points.	4	K3

7. Attempt any *one* part of the following: **07 x 1 = 07**

Qno.	Question	CO	Level
a.	Explain MapReduce model with architecture and working in Apache Hadoop.	5	K2
b.	Explain Exploratory Data Analysis (EDA) in R with steps and sample commands.	5	K2

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